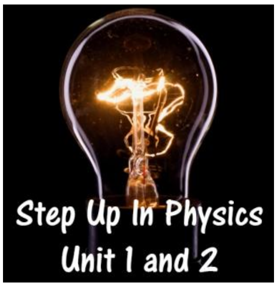
Thermal Equilibrium

Problems Worksheet



1. Describe the conditions that are required for two mixed liquids to be in thermal equilibrium.

2. 350 ml of hot water at 85.0 ^{0}C is mixed with 120 ml of cool water at 12.0 ^{0}C in a plastic cup. Determine the final temperature of the mixture.

3. Calculate the final temperature of 0.180 kg of ice at 0.00 ^{0}C added to an insulated container filled with 4.50 kg of water at 20.0 ^{0}C.

4. Calculate the final temperature of 10.0 g of ice at -15.0 ^{0}C added to a styrofoam cup filled with 220 ml of water at 32.0 ^{0}C.

5. Two identical beakers each contain the same quantity of water at 60.0 ^{0}C. 10.0 g of ice at its melting point is added to one beaker. The other beaker has 10.0 g of water at 0.00 ^{0}C added. Explain which beaker will have the lowest temperature once thermal equilibrium is reached.

6. An unknown metal weighing 900 g at an initial temperature of 140 ^{0}C is placed into an insulated container holding 3.00 L of water at an initial temperature of 60.0 ^{0}C. The water rose to 65.0 ^{0}C. Calculate the specific heat capacity of the metal.

7. Calculate the mass of ice at 0 ^{0}C needed to cool 150 ml of water from 60.0 °C to 40.0 ^{0}C if the water is in a 100 g copper cup. Copper has a specific heat capacity of 390 J kg^{-1} K^{-1}.

8. Calculate the mass of water at 32.0 ^{0}C needed to melt 1.50 kg of ice at -10.0 ^{0}C.

9. A copper cup holds some cold water at 4.00 ^{0}C. The copper cup has a mass of 140 g while the water has a mass of 80.0 g. If 100 g of hot water, at 90.0 ^{0}C is added to the cup, what will the final temperature of the water mixture be? Copper has a specific heat capacity of 390 J kg^{-1} K^{-1}.

10. 3.00 kg of ice at 0.00 ^{0}C was completely melted by 250 g of steam. Assuming the melted ice remained at 0.00 ^{0}C, calculate the initial temperature of the steam.

11. Calculate the maximum mass of ice at -10.0 ^{0}C that can be melted by 180 g of steam at 115 ^{0}C.

12. A 20.0 g piece of lead at 25.0 ^{0}C is placed into a furnace that transfers 1.16×10^{3} J of thermal energy into the lead. Using the information in the table below, calculate how much lead remains unmelted in the furnace.

Variable	**Value**
Specific heat capacity of lead	128 J kg^{-1} K^{-1}
Latent heat of fusion of lead	2.30×10^{4} J kg^{-1}
Latent heat of vaporisation of lead	8.71×10^{5} J kg^{-1}
Melting point of lead	327 ^{0}C

13. During an experiment to determine the specific heat capacity of a brick, two students heated a brick in an oven until they were sure it was at 220 ^{0}C. They removed the brick from the oven using oven mitts, took it outside to where they had a 9.00 L bucket that was 95% filled with water and dropped the brick into the water.

    a. List the measurements not already mentioned that the students would need to make to determine the specific heat capacity of the brick.

    b. List two major sources of error in their experiment that would affect their determination of the specific heat capacity of the brick.

    c. For both major sources of error, justify whether this would make their calculation of the specific heat capacity of the brick higher or lower than the expected value.

